BINSTD runner source restoration with prior toy replay

Validation audit, 22 October 2026

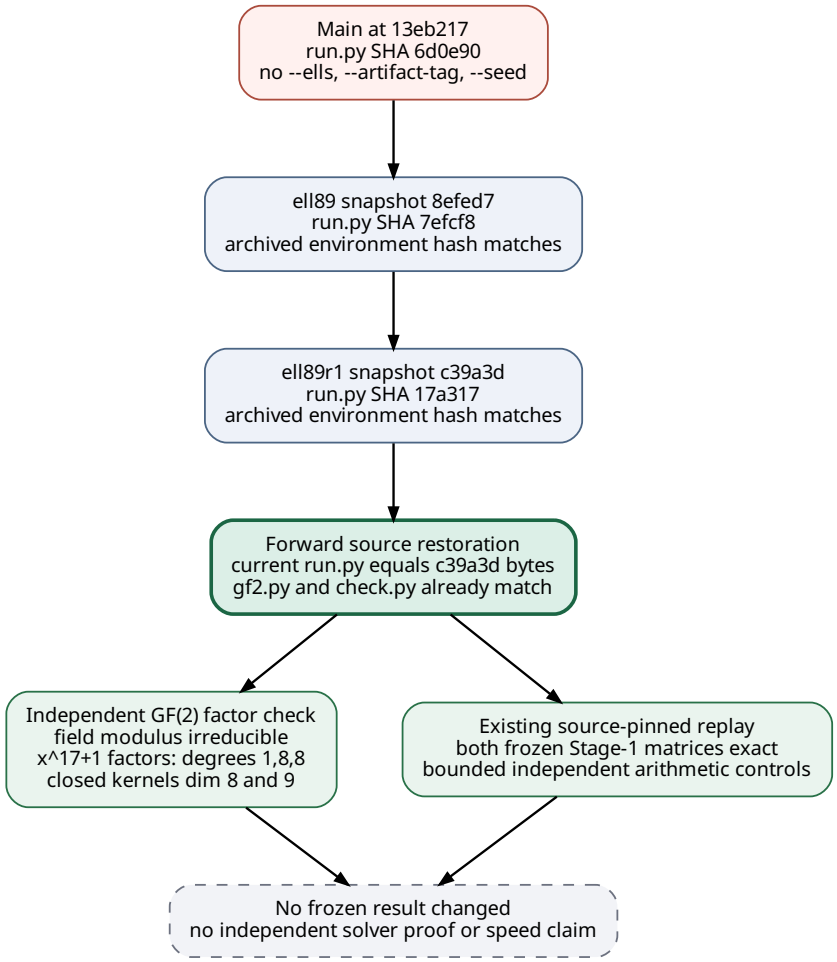
Finding. On main `13eb21767865`, the BINSTD runner accepts none of the `-ells`, `-artifact-tag` and `-seed` options used by the archived Stage-1 commands. The current runner's SHA-256 begins `6d0e90f863c4`. The ell89 snapshot at `8efed7dea288` contains source hash `7efcf8e6bbfc`; the ell89r1 snapshot at `c39a3d5e4cc7` contains `17a3175a30b2`. Each matches the corresponding frozen run environment. The field helper and checker match both snapshots and current main.

Correction. The live `implementation/run.py` is now the exact `c39a3d5e4cc7` runner. This restores both additive panels and their CLI without altering a run or scientific record. The source-custody receipt binds the snapshots, run environments and restored file.

Algebra check. An independent SymPy factorization confirms that $t^{17}+t^3+1$ is irreducible over $\mathbf{F}_2$ and that $x^{17}+1$ has irreducible degrees $1, 8, 8$. The restored code builds Frobenius-closed kernels of dimensions 8 and 9; dimensions 3 and 4 are excluded. Sampled curve equations, group associativity and small-factor-base triple membership pass. The full ledger validator still exits 1 with the same 141 errors in 39 paths as main; the log is byte-for-byte unchanged.

Prior replay. An existing source-pinned audit already ran both archived Stage-1 panels in temporary directories. Its receipt records exact matches for both frozen dimension tables, yield matrices and result notes, plus bounded independent arithmetic controls. This patch restores that latest historical source to current main; the completed full-panel replay was not repeated.

Boundary. The prior deterministic replay of the archived toy algorithm is not an independent per-target relation proof or a Lean yield theorem. It does not change the measured ratios, scoped decisions, performance estimates or any deployed-curve claim.


Main at 13eb217
run.py SHA 6d0e90
no --ells, --artifact-tag, --seed
ell89 snapshot 8efed7
run.py SHA 7efcf8
archived environment hash matches
ell89r1 snapshot c39a3d
run.py SHA 17a317
archived environment hash matches
Forward source restoration
current run.py equals c39a3d bytes
gf2.py and check.py already match
Independent GF(2) factor check
field modulus irreducible
x^17+1 factors: degrees 1,8,8
closed kernels dim 8 and 9
Existing source-pinned replay
both frozen Stage-1 matrices exact
bounded independent arithmetic controls
No frozen result changed
no independent solver proof or speed claim